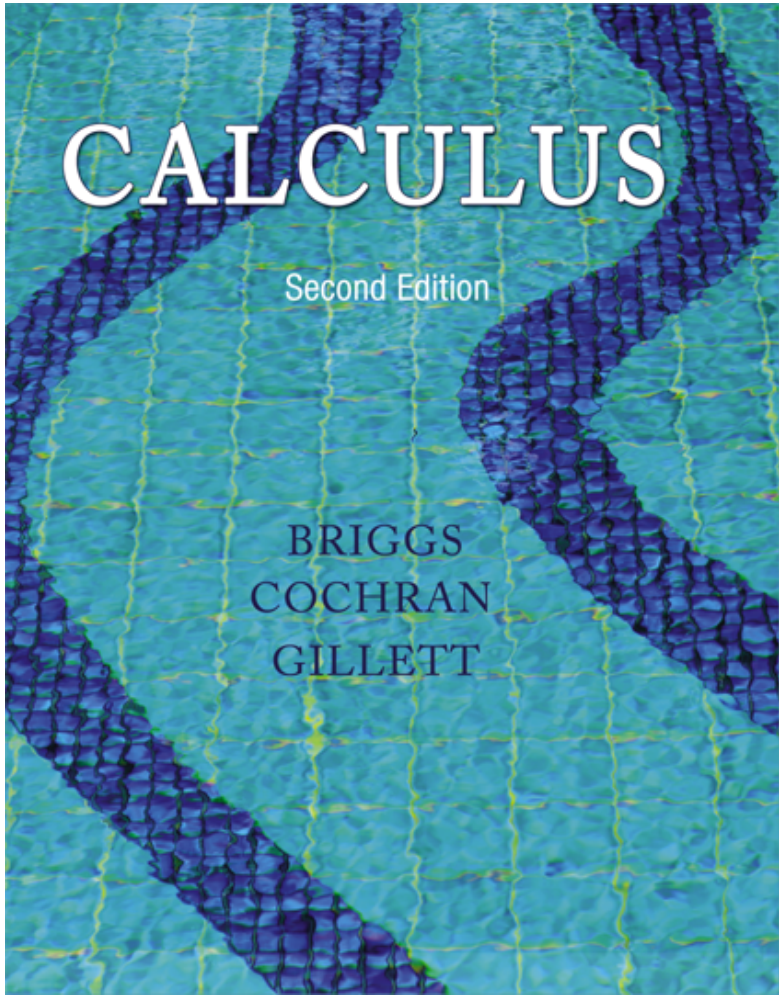
Chapter 9

Sequences and Infinite Series



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9.2

Sequences

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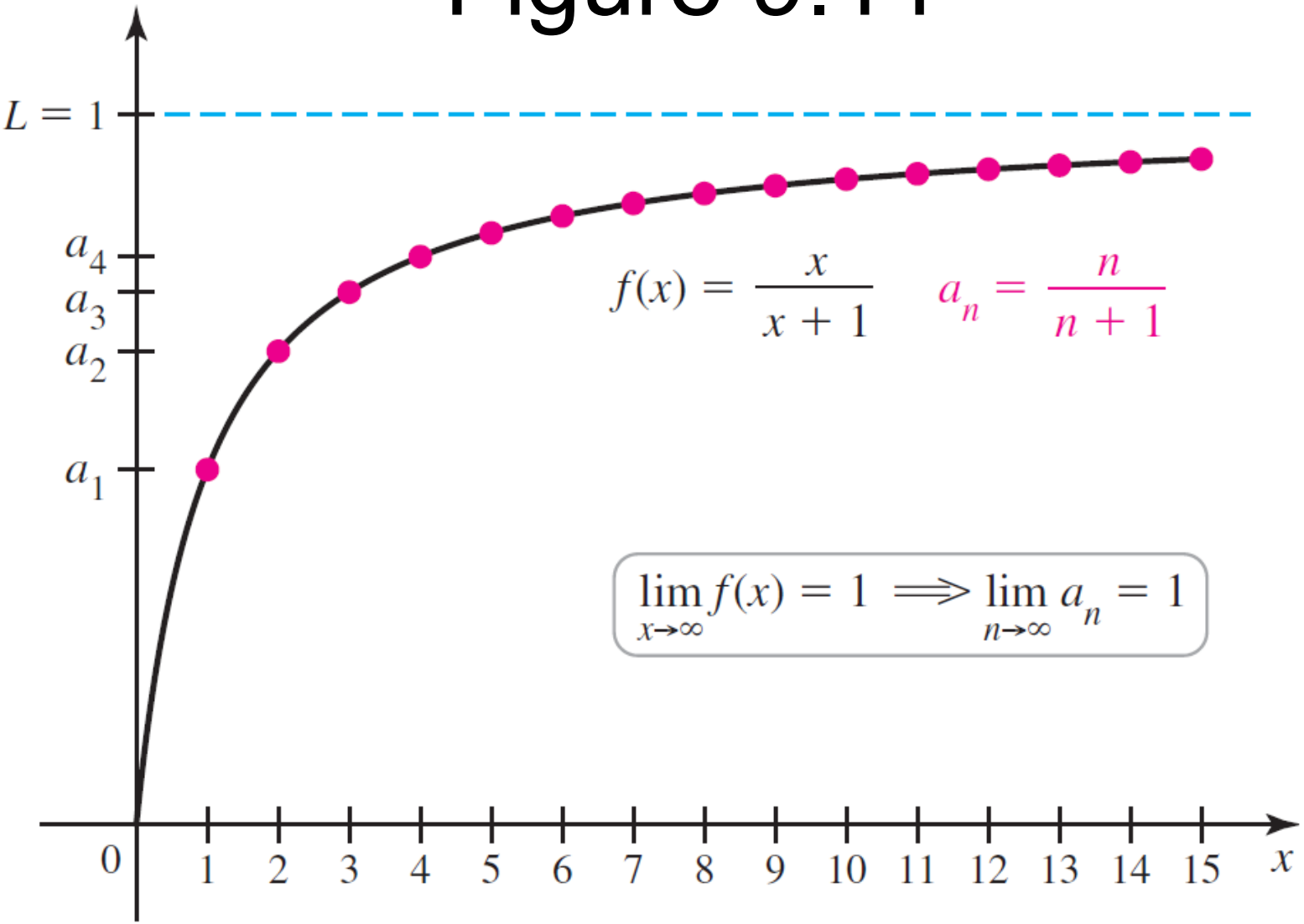
Figure 9.11



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THEOREM 9.1 Limits of Sequences from Limits of Functions

Suppose f is a function such that $f(n) = a_n$ for all positive integers n. If $\lim_{x \to \infty} f(x) = L$, then the limit of the sequence $\{a_n\}$ is also L.

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THEOREM 9.2 Limit Laws for Sequences

Assume that the sequences $\{a_n\}$ and $\{b_n\}$ have limits A and B, respectively. Then

1. $\lim_{n\to\infty} (a_n \pm b_n) = A \pm B$
2. $\lim_{n\to\infty} ca_n = cA$, where c is a real number
3. $\lim_{n\to\infty} a_n b_n = AB$
4. $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{A}{B}$, provided $B \neq 0$.

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DEFINITIONS Terminology for Sequences

$\{a_n\}$ is **increasing** if $a_{n+1} > a_n$; for example, $\{0, 1, 2, 3, \ldots\}$.

$\{a_n\}$ is **nondecreasing** if $a_{n+1} \geq a_n$; for example, $\{1, 1, 2, 2, 3, 3, \ldots\}$.

$\{a_n\}$ is **decreasing** if $a_{n+1} < a_n$; for example, $\{2, 1, 0, -1, \ldots\}$.

$\{a_n\}$ is **nonincreasing** if $a_{n+1} \leq a_n$; for example, $\{0, -1, -1, -2, -2, -3, -3, \ldots\}$.

$\{a_n\}$ is **monotonic** if it is either nonincreasing or nondecreasing (it moves in one direction).

$\{a_n\}$ is **bounded** if there is number M such that $|a_n| \leq M$, for all relevant values of n.

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Figure 9.12

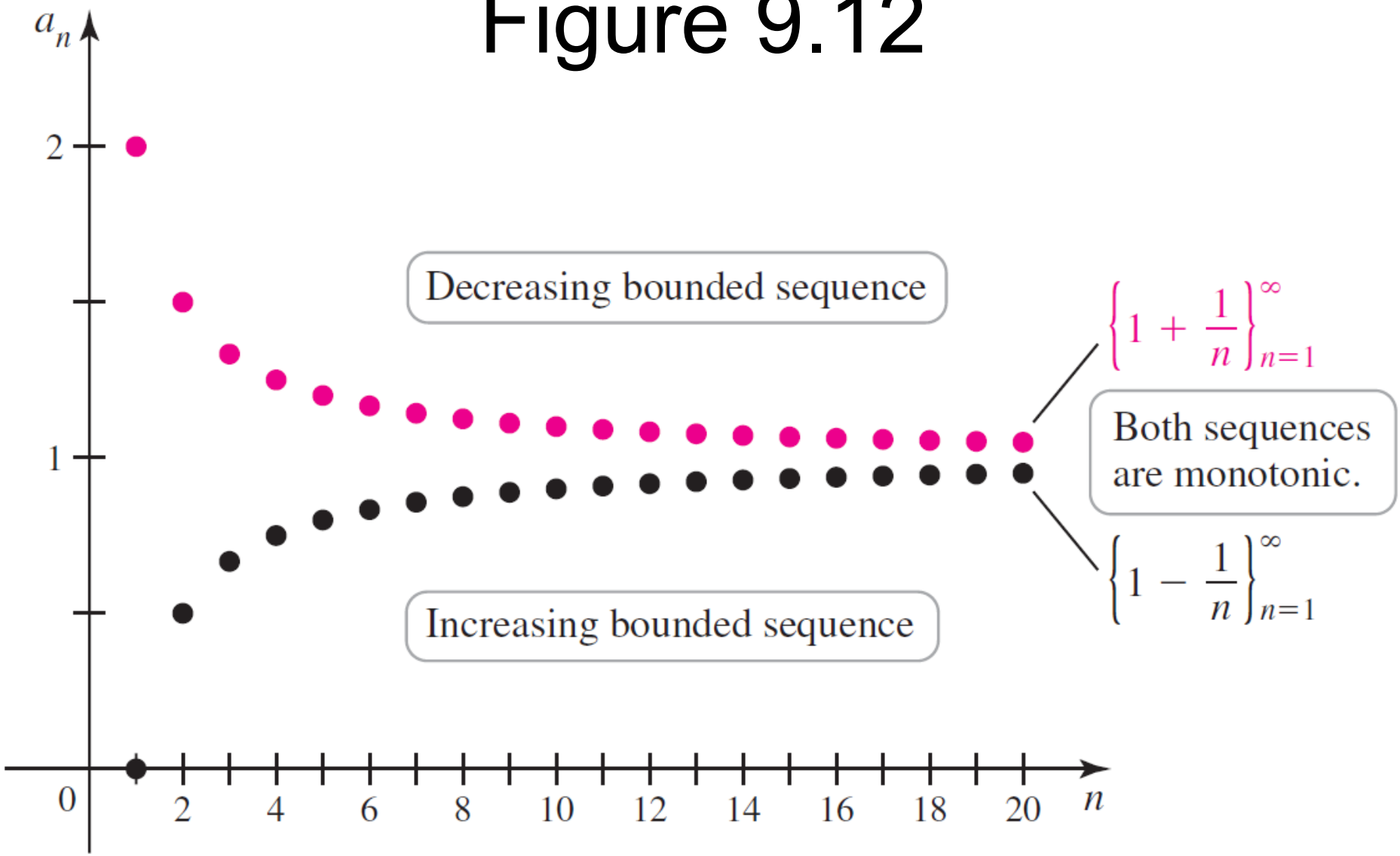

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Figure 9.13

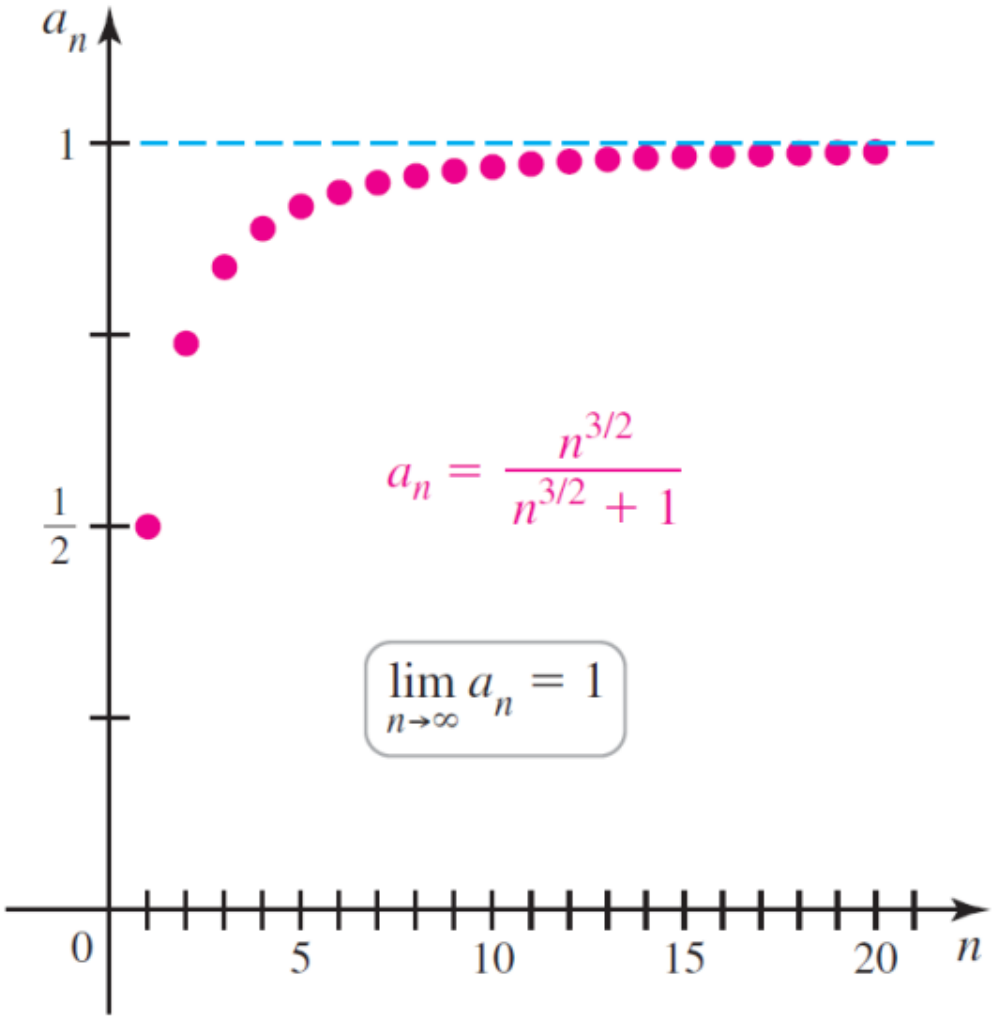

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Figure 9.14

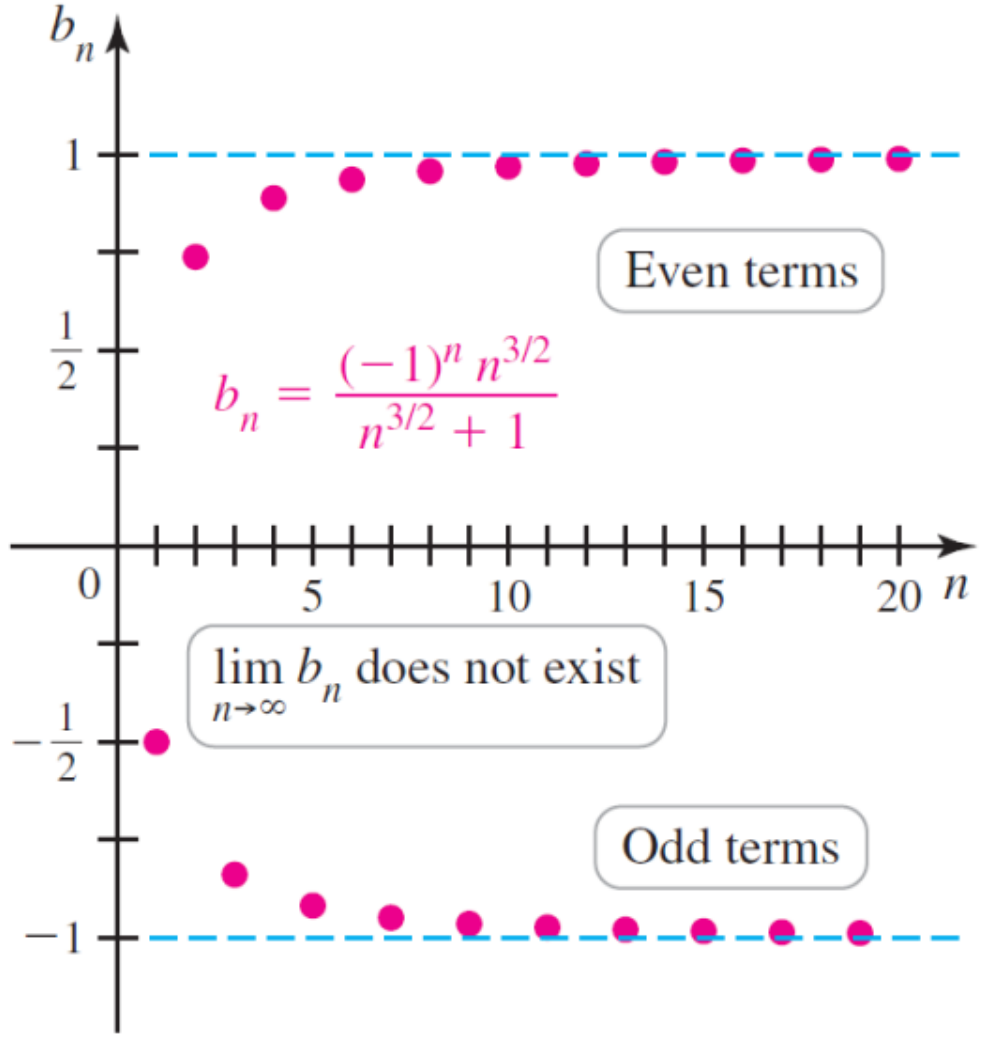

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Figure 9.15

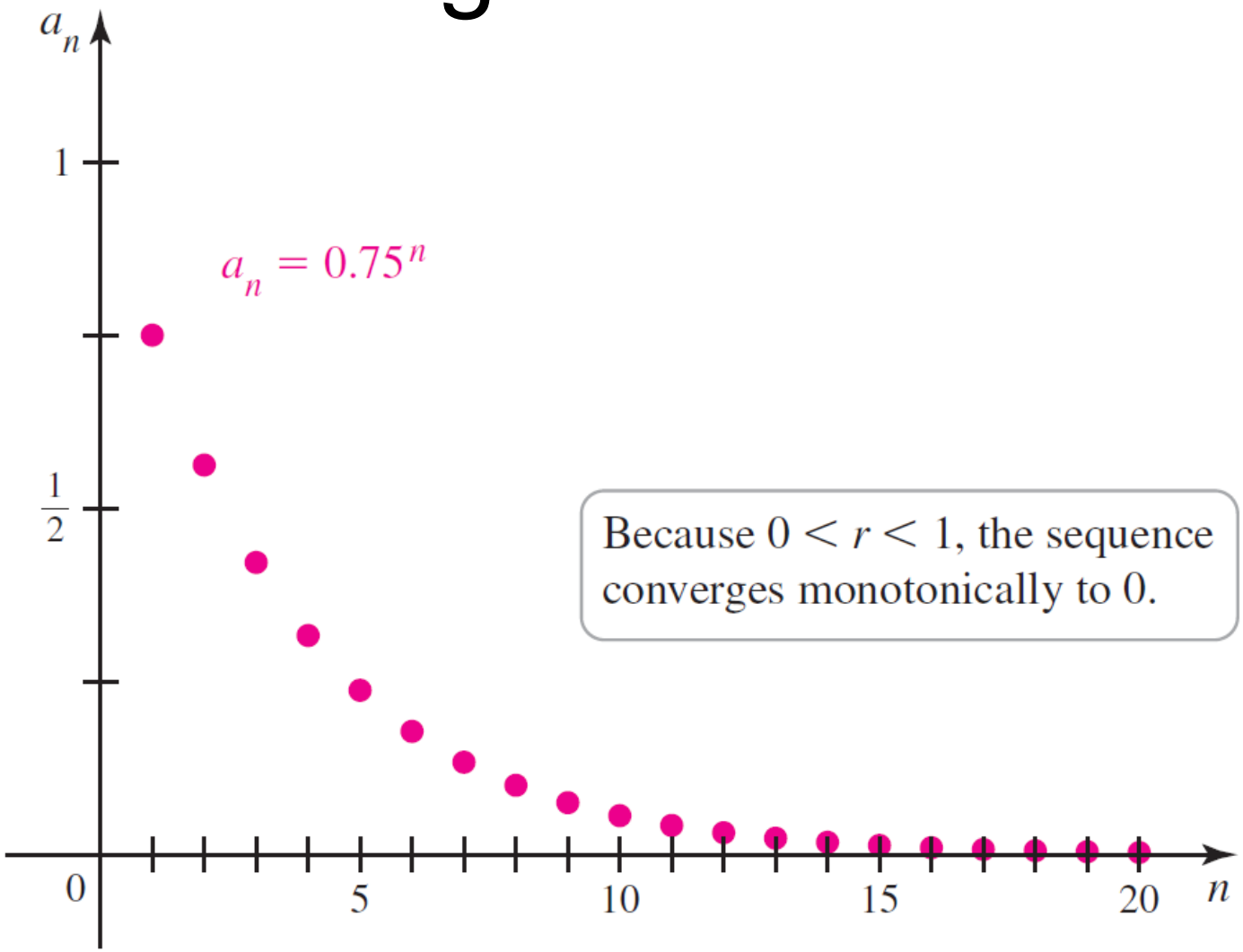

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Figure 9.16

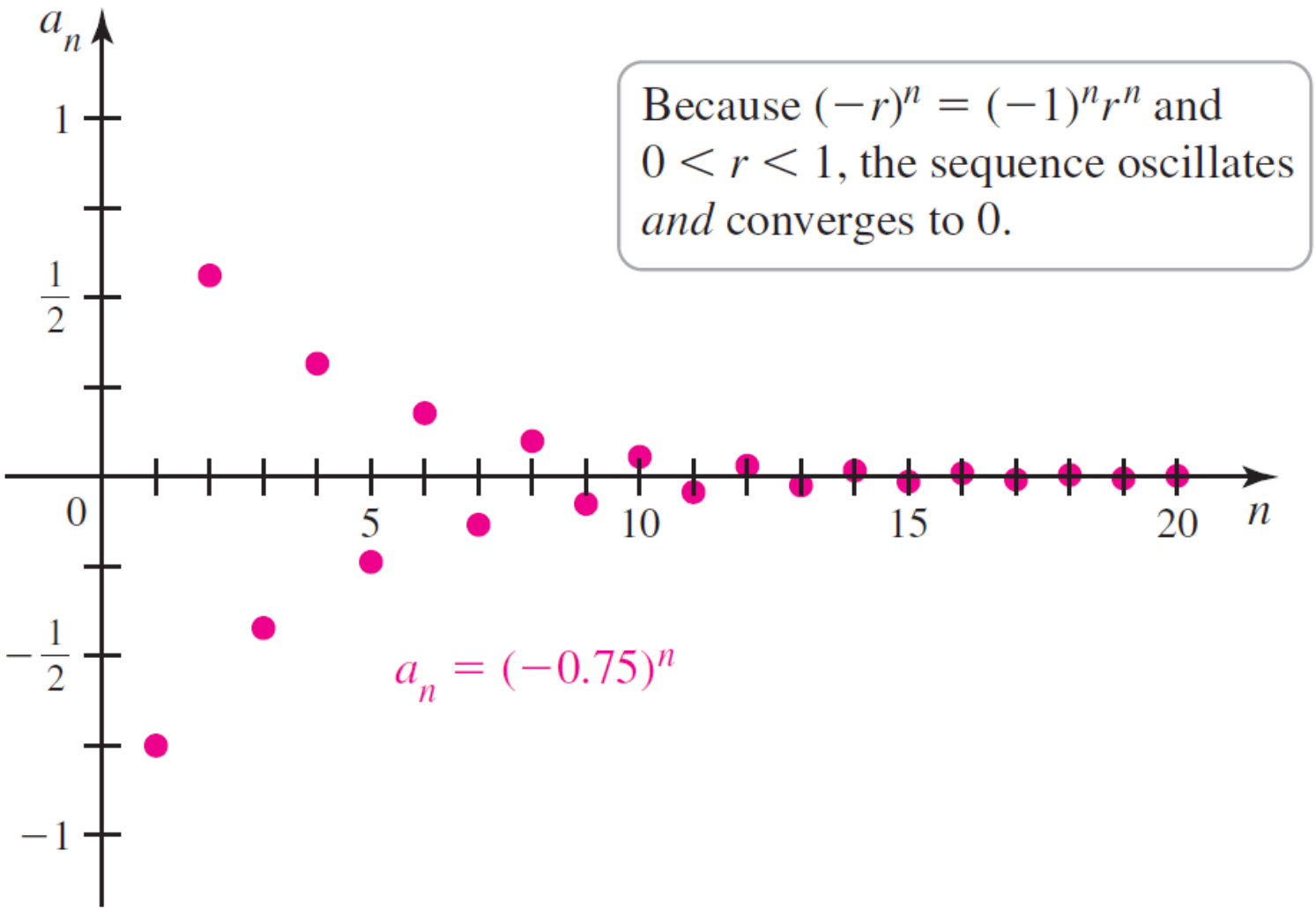

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Figure 9.17

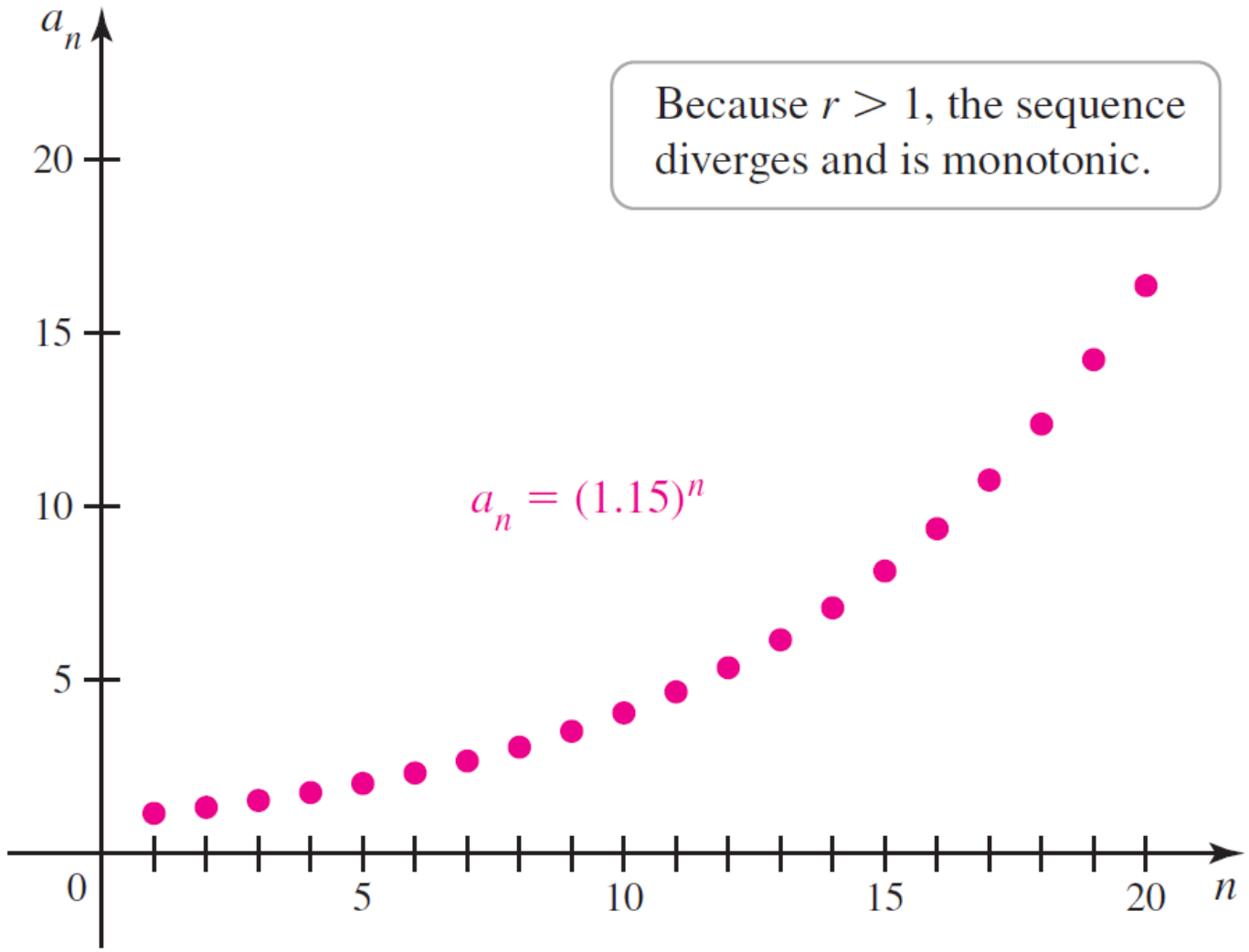

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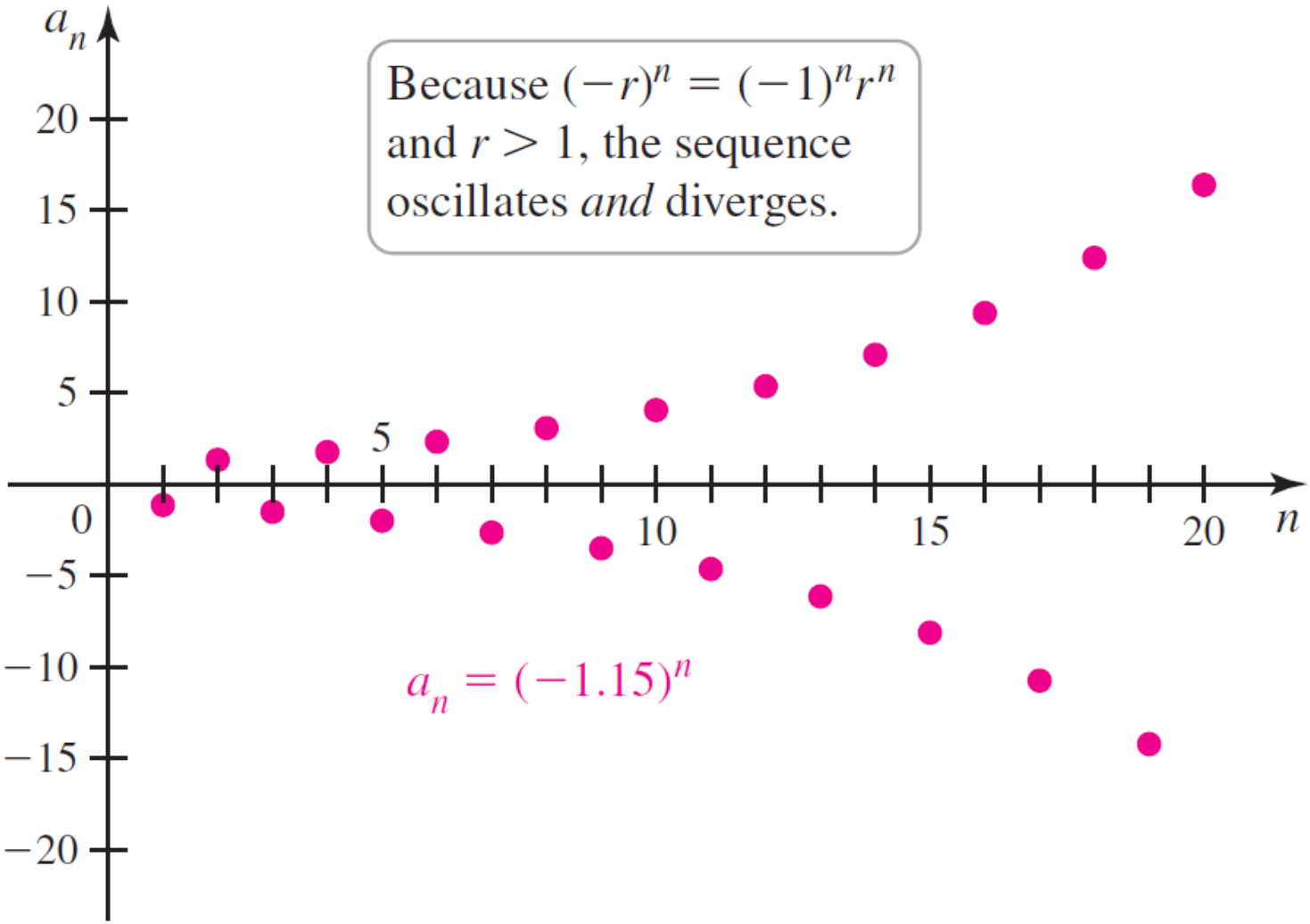
Figure 9.18



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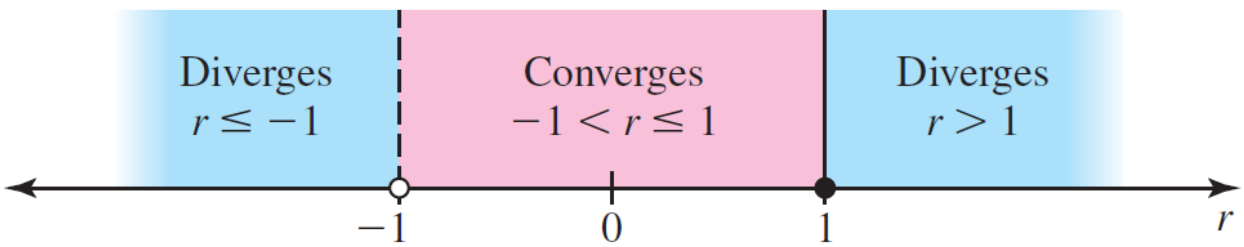
THEOREM 9.3 Geometric Sequences

Let r be a real number. Then

$$\lim_{n\to\infty} r^n = \begin{cases} 0 & \text{if } |r| < 1 \\ 1 & \text{if } r = 1 \\ \text{does not exist} & \text{if } r \le -1 \text{ or } r > 1. \end{cases}$$

If $r > 0$, then $\{r^n\}$ is a monotonic sequence. If $r < 0$, then $\{r^n\}$ oscillates.



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THEOREM 9.4 Squeeze Theorem for Sequences

Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ be sequences with $a_n \le b_n \le c_n$ for all integers n greater than some index N. If $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, then $\lim_{n\to\infty} b_n = L$ (Figure 9.19).

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Figure 9.19

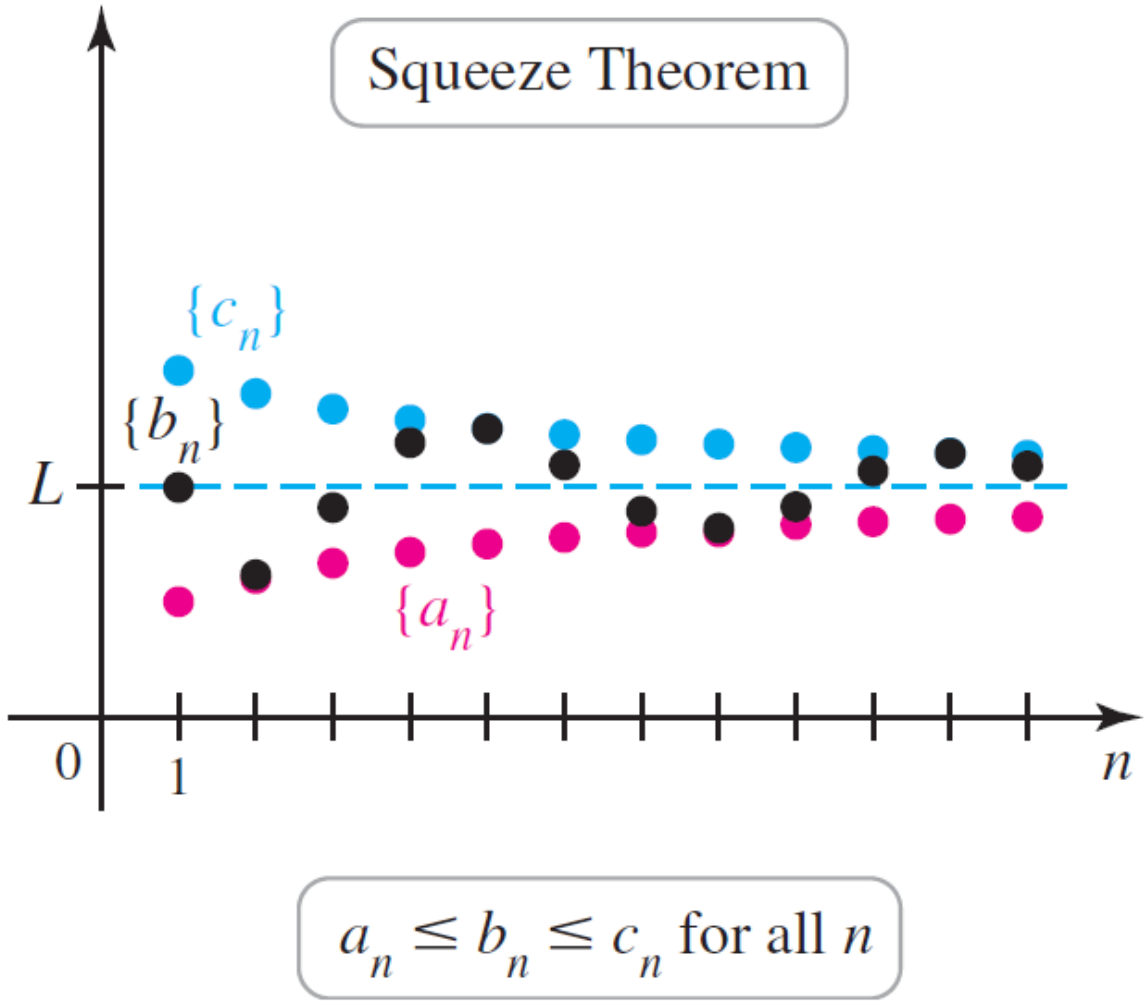

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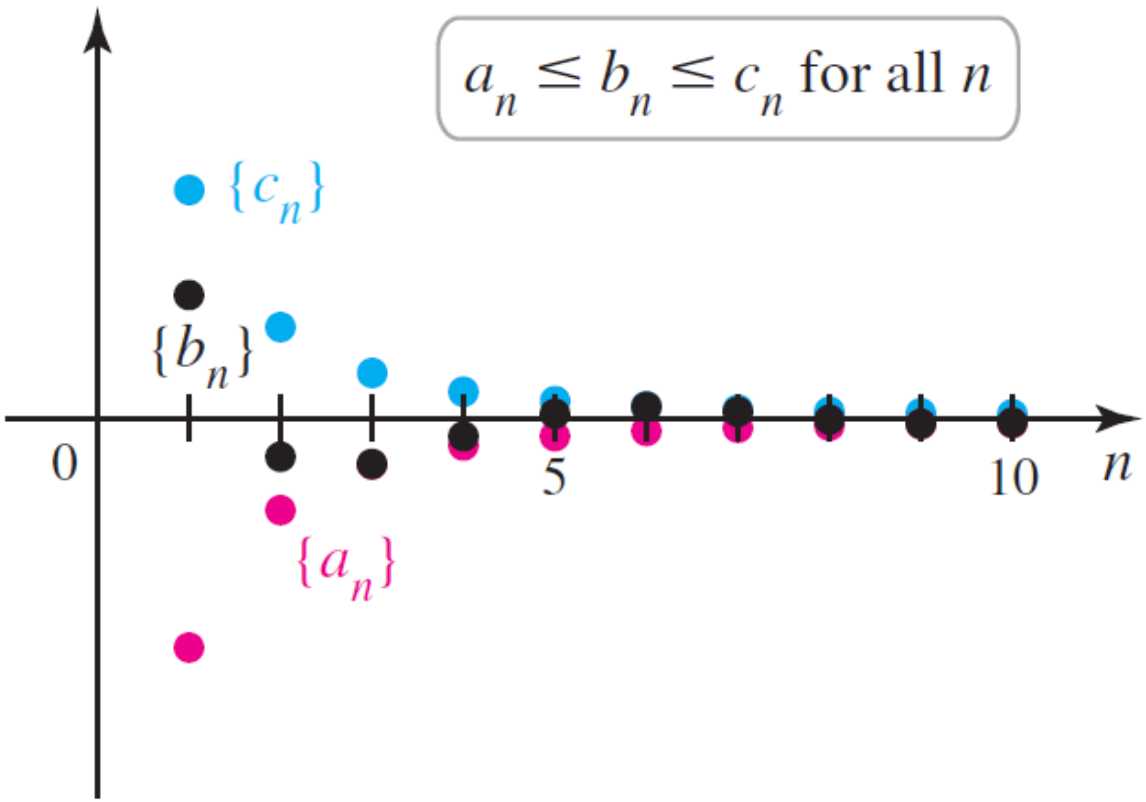
Figure 9.20



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THEOREM 9.5 Bounded Monotonic Sequences

A bounded monotonic sequence converges.

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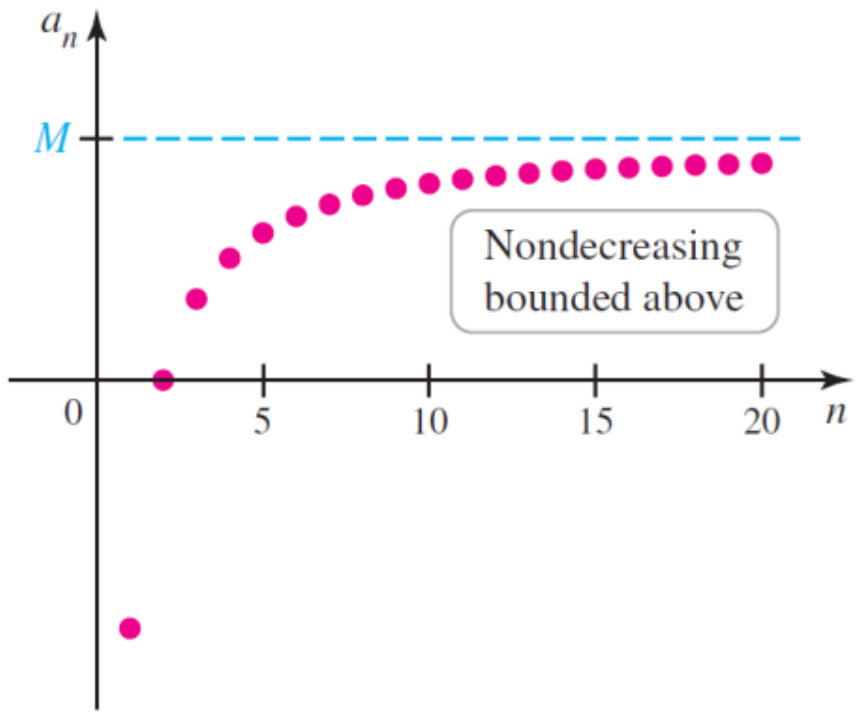
Figure 9.21 (a & b)



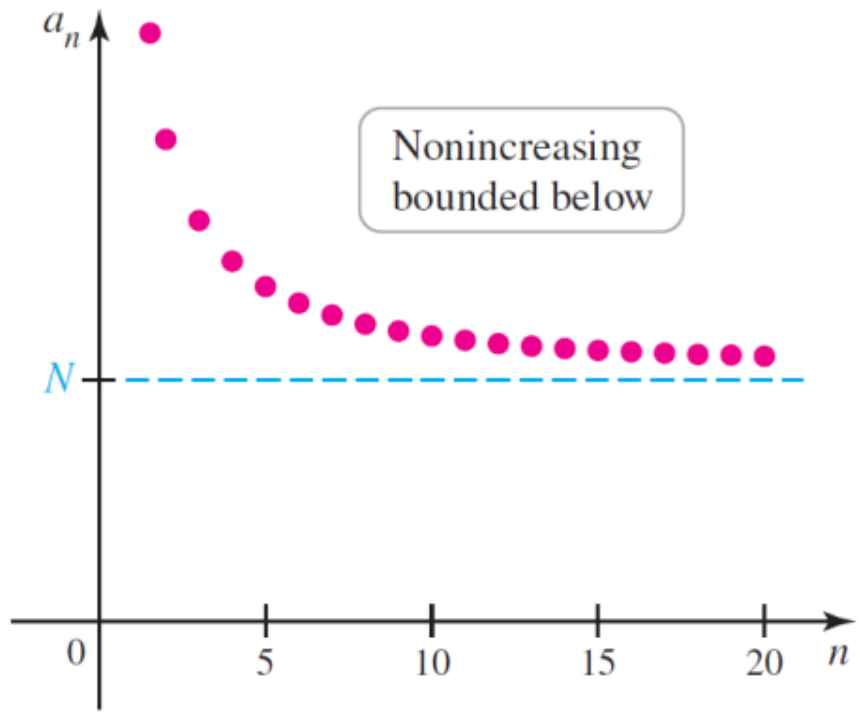


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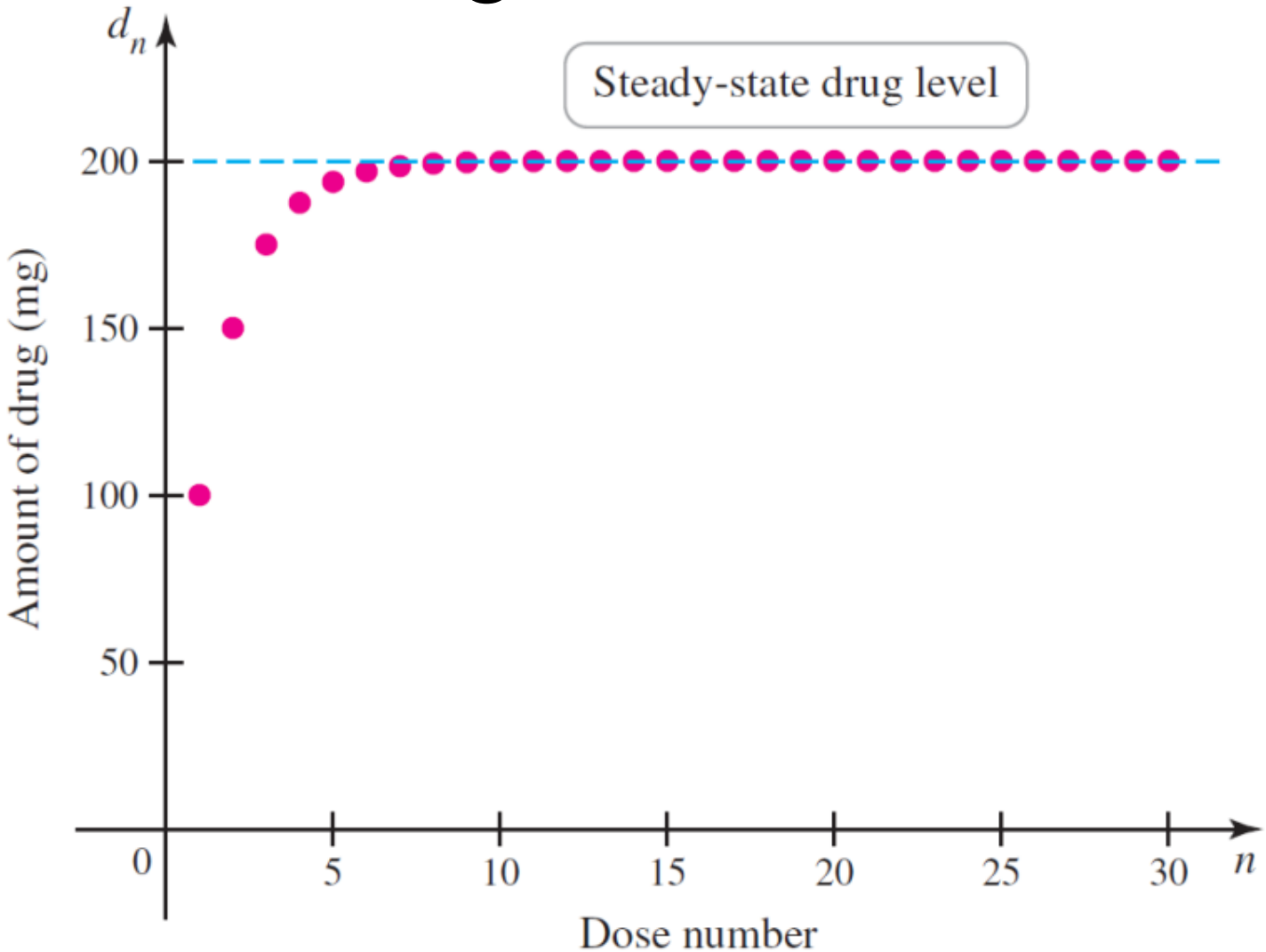
Figure 9.22



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THEOREM 9.6 Growth Rates of Sequences

The following sequences are ordered according to increasing growth rates as $n \to \infty$; that is, if $\{a_n\}$ appears before $\{b_n\}$ in the list, then $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ and $\lim_{n\to\infty} \frac{b_n}{a_n} = \infty$:

$$\{\ln^q n\} \ll \{n^p\} \ll \{n^p \ln^r n\} \ll \{n^{p+s}\} \ll \{b^n\} \ll \{n!\} \ll \{n^n\}.$$

The ordering applies for positive real numbers p, q, r, s, and $b > 1$.

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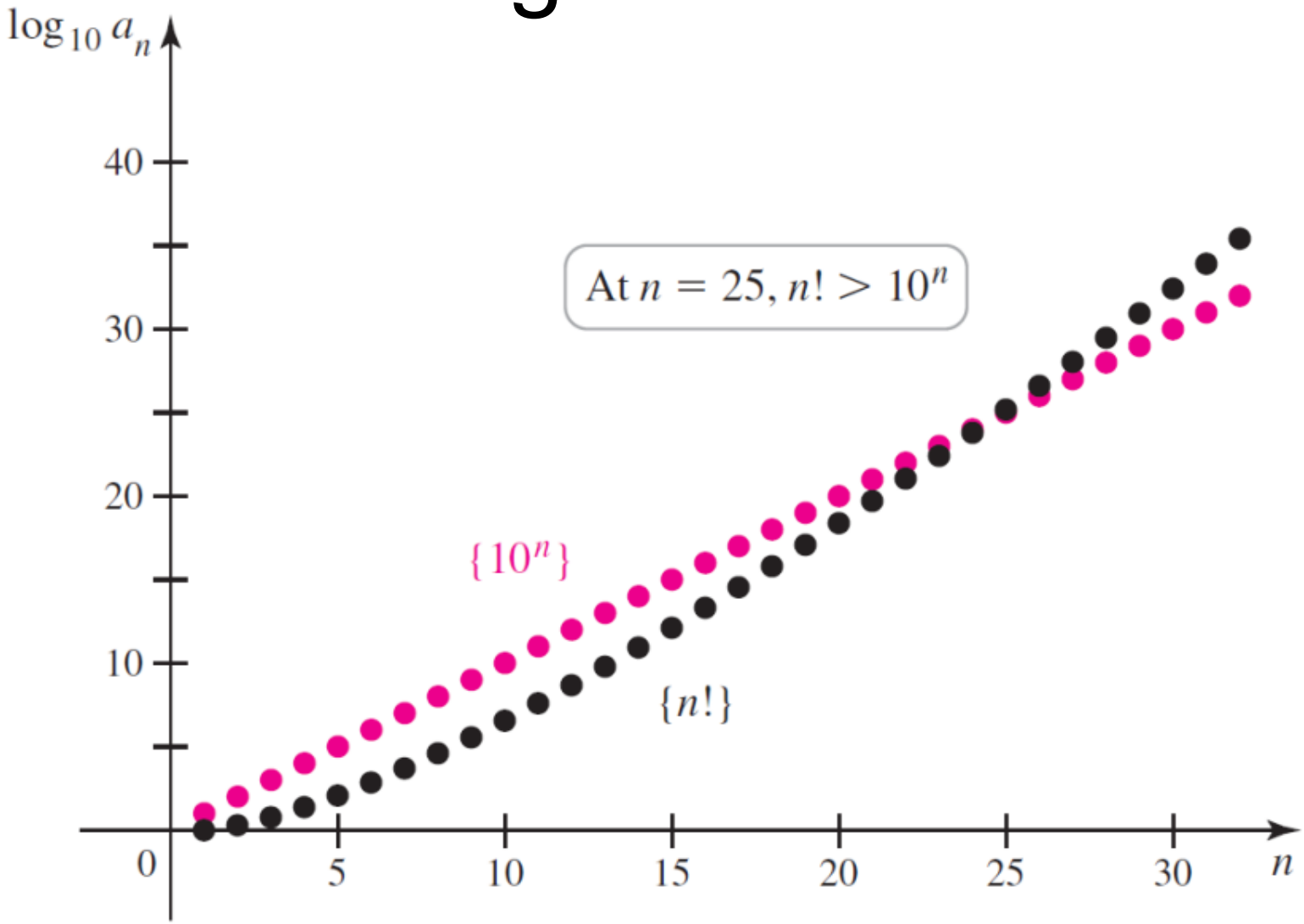
Figure 9.23



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DEFINITION Limit of a Sequence

The sequence $\{a_n\}$ converges to L provided the terms of a_n can be made arbitrarily close to L by taking n sufficiently large. More precisely, $\{a_n\}$ has the unique limit L if given any $\varepsilon > 0$, it is possible to find a positive integer N (depending only on ε) such that

$$|a_n - L| < \varepsilon \qquad \text{whenever } n > N.$$

If the **limit of a sequence** is L, we say the sequence **converges** to L, written

$$\lim_{n \to \infty} a_n = L.$$

A sequence that does not converge is said to **diverge**.

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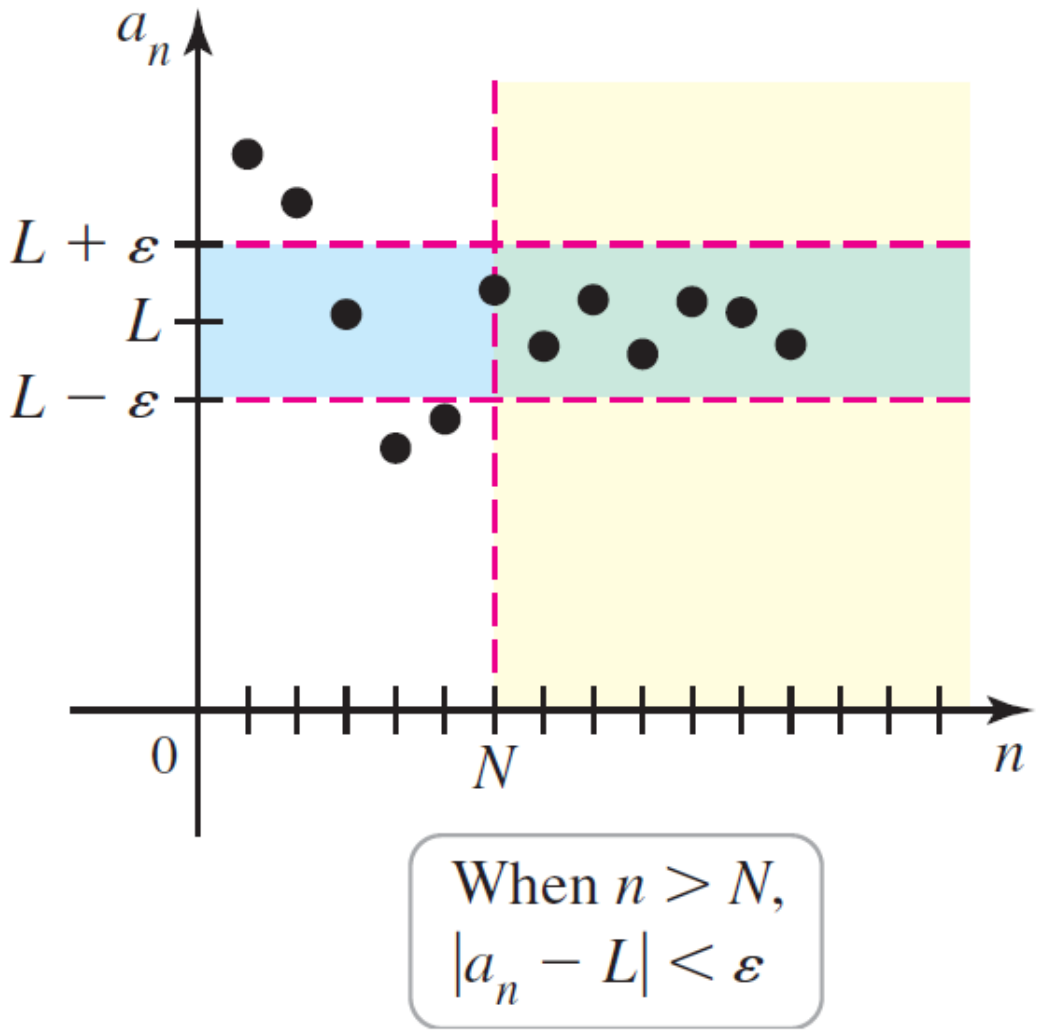
Figure 9.24



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